Determine the period of the following function

$$f(x) = 1952 \cos^{1971}(2025x + 2026)$$

Solution:

period $(\cos x) = 2\pi$

" $(\cos^{1971} x) = 2\pi$

" $(f(x)) = \dfrac{2\pi}{2025}$

1972

$\cos^{1972} x = \pi$

$\cos^{1972}(2025x) = \dfrac{\pi}{2025}$

Sketch the periodin function:

(a) $f(x) = |x| \; ; \; -\pi < x < \pi$

(b) $f(x) = 2\sin x \; ; \; 0 < x < \pi$

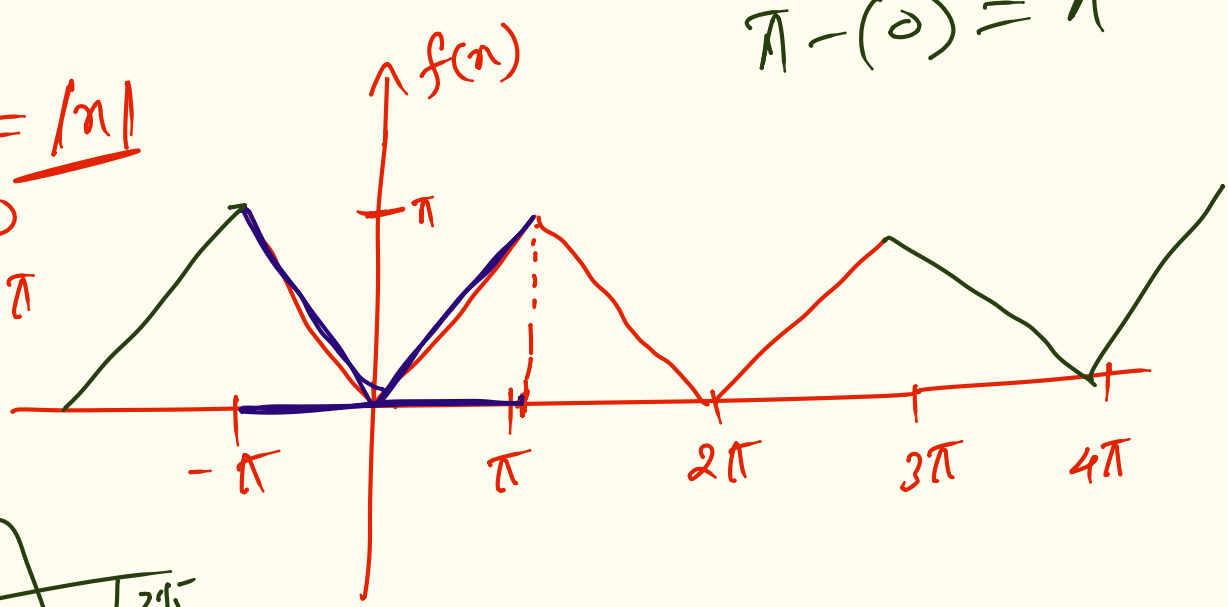
$\pi - (0) = \pi$

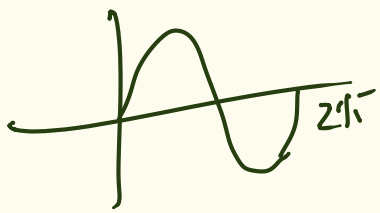
$f(x) = |x|$

$f(0) = 0$

$f(-\pi) = \pi$

$f(\pi) = \pi$

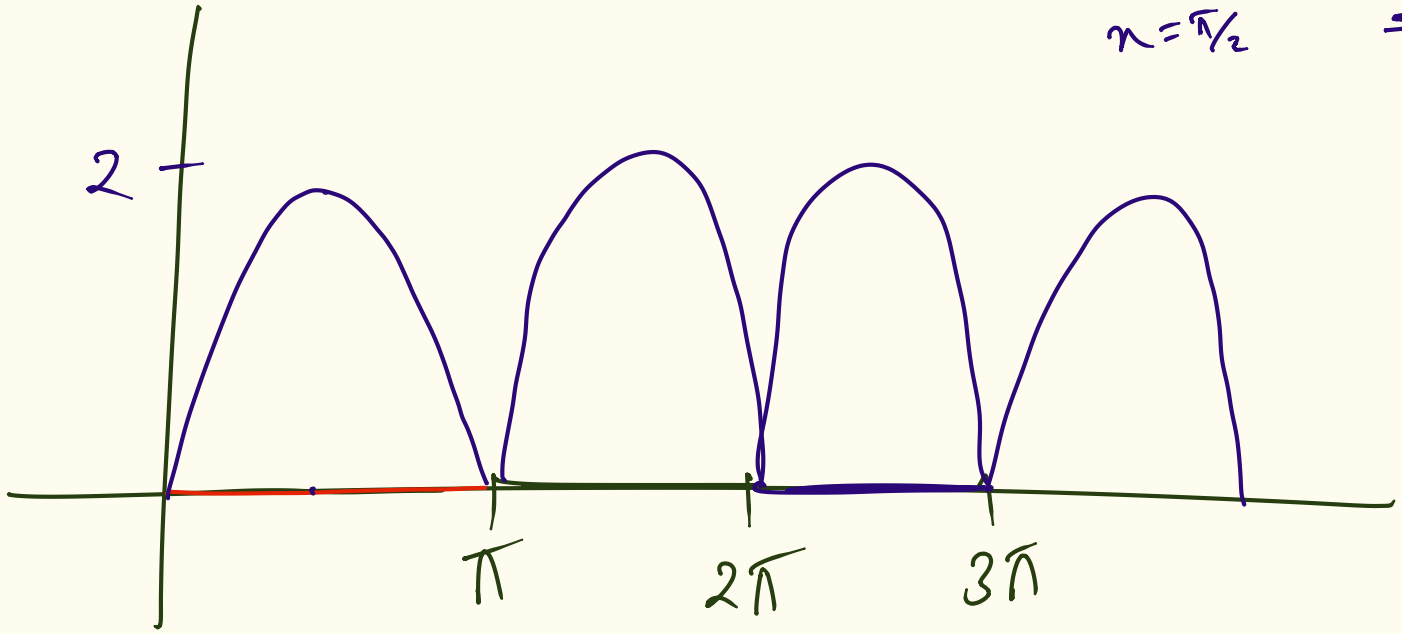




$x = 0 \quad f(0) = 0$

$x = \pi/2 \quad = 2 \cdot 1$

$= 2$



Fourier Series

$f(x)$ is a periodic function on $-L < x < L$ with period $= 2L$ then,

$$f(x) = \frac{a_0}{2} + \sum_{n=1}^{\infty}\left[a_n \cos\frac{n\pi x}{L} + b_n \sin\frac{n\pi x}{L}\right]; \quad -L < x < L \qquad (1)$$

Finding a_0

Now integrate (1) from $-L$ to L w.r.to x.

$$\int_{-L}^{L} f(x)\,dx = \int_{-L}^{L} \frac{a_0}{2}\,dx + \sum_{n=1}^{n}\int_{-L}^{L} a_n \cos\frac{n\pi x}{L} + b_n \sin\frac{n\pi x}{L}\,dx$$

Here

$$\int_{-L}^{L} \cos\frac{n\pi x}{L}\,dx = \frac{L}{n\pi}\left[\sin\frac{n\pi x}{L}\right]_{-L}^{L}$$

$$= \frac{L}{n\pi}\left[\sin\frac{n\pi L}{L} - \sin\frac{n\pi(-L)}{L}\right]$$

$\sin(-\theta) = -\sin\theta$

$$= \frac{L}{n\pi}\left[\sin(n\pi) - \sin(-n\pi)\right]$$

$$= \frac{L}{n\pi}\left[\sin(n\pi) + \sin(n\pi)\right]$$

$$= \frac{L}{n\pi}(0+0) = 0$$

Here,

$$\int_{-L}^{L} \sin\frac{n\pi x}{L}\,dx = \frac{-L}{n\pi}\left[\cos\frac{n\pi x}{L}\right]_{-L}^{L}$$

$$= \frac{-L}{n\pi}\left[\cos\frac{n\pi L}{L} - \cos\frac{n\pi(-L)}{L}\right]$$

$\cos(-\theta) = \cos\theta$

$\cos(n\pi) = (-1)^n$

$$= \frac{-L}{n\pi}\left[\cos(n\pi) - \cos(-n\pi)\right]$$

$$= \frac{-L}{n\pi}\left[\cancel{\cos(n\pi)} - \cancel{\cos(n\pi)}\right]$$

$$= 0$$

$$\int_{-L}^{L} f(x)\,dx = \int_{-L}^{L} \frac{a_0}{2}\,dx + 0 + 0$$

$$= \frac{a_0}{2}\left[x\right]_{-L}^{L} = \frac{a_0}{2}\left(L - (-L)\right)$$

$$\Rightarrow \int_{-L}^{L} f(x)\,dx = \frac{a_0}{2}\,2L$$

$$\boxed{a_0 = \frac{1}{L}\int_{-L}^{L} f(x)\,dx}$$

$$f(x) = \frac{a_0}{2} + \sum_{n=1}^{\infty}\left[a_n\cos\frac{n\pi x}{L} + b_n\sin\frac{n\pi x}{L}\right]; \quad -L < x < L \qquad (1)$$

① $\times \cos\frac{n\pi x}{L}$ and integrate

$$\int_{-L}^{L} f(x)\cos\frac{n\pi x}{L}\,dx = \int_{-L}^{L}\frac{a_0}{2}\cos\frac{n\pi x}{L}\,dx$$

$$+\sum_{n=1}^{\infty}\int_{-L}^{L} a_n\cos\frac{n\pi x}{L}\cdot\cos\frac{n\pi x}{L}\,dx$$

$$+\sum_{n=1}^{\infty}\int_{-L}^{L} b_n\sin\frac{n\pi x}{L}\cdot\cos\frac{n\pi x}{L}\,dx$$

$$= 0 + a_n L + 0$$

$$\Rightarrow \quad a_n = \frac{1}{L}\int_{-L}^{L} f(x)\cos\frac{n\pi x}{L}\,dx$$

Again, ① $\times \sin\frac{n\pi x}{L}$ and integrate

$$b_n = \frac{1}{L}\int_{-L}^{L} f(x)\sin\left(\frac{n\pi x}{L}\right)dx$$

$$f(x) = \frac{a_0}{2} + \sum_{n=1}^{\infty}\left[a_n \cos\frac{n\pi x}{L} + b_n \sin\frac{n\pi x}{L}\right]; \quad -L < x < L \qquad (1)$$

$$a_0 = \frac{1}{L}\int_{-L}^{L} f(x)\,dx$$

$$a_n = \frac{1}{L}\int_{-L}^{L} f(x)\cos\frac{n\pi x}{L}\,dx$$

$$b_n = \frac{1}{L}\int_{-L}^{L} f(x)\sin\left(\frac{n\pi x}{L}\right)dx$$

Find the Fourier series expansion of

$$f(x) = \pi - x \text{ on } (-\pi, \pi)$$

Solution: Here period $\pi - (-\pi) = 2\pi$

$$\therefore 2L = 2\pi \Rightarrow \boxed{L = \pi}$$

$$a_0 = \frac{1}{L}\int_{-L}^{L} f(x)\,dx$$

$$= \frac{1}{\pi}\int_{-\pi}^{\pi}(\pi - x)\,dx$$

$$= \frac{1}{\pi}\left[\pi x - \frac{x^2}{2}\right]_{-\pi}^{\pi} =$$

$$= \frac{1}{\pi}\left[\left(\pi^2 - \cancel{\frac{\pi^2}{2}} - (-\pi^2) + \cancel{\frac{\pi^2}{2}}\right)\right]$$

$$= \frac{1}{\pi}\, 2\pi^2 = 2\pi$$

$$\boxed{\therefore a_0 = 2\pi}$$

$$a_n = \frac{1}{L}\int_{-L}^{L} f(x)\cos\frac{n\pi x}{L}\,dx$$

$$= \frac{1}{\pi}\int_{-\pi}^{\pi}(\pi - x)\cos\frac{n\pi x}{\pi}\,dx$$

u	v
$(+)\ (\pi - x)$	$\cos nx$
$(-)\ -1$	$\frac{\sin nx}{n}$
0	$\frac{-\cos nx}{n^2}$

$$= \frac{1}{\pi}\int_{-\pi}^{\pi}(\pi - x)\cos nx\,dx$$

$$= \frac{1}{\pi}\left[(\pi - x)\frac{\sin(nx)}{n} - \frac{\cos(nx)}{n^2}\right]_{-\pi}^{\pi}$$

$$= \frac{1}{\pi}\left[\left\{(\pi-\pi)\frac{\sin(n\pi)}{n} - \frac{\cos(n\pi)}{n^2}\right\} - \left\{(\pi+\pi)\frac{\sin(-n\pi)}{n} - \frac{\cos(-n\pi)}{n^2}\right\}\right]$$

$$= \frac{1}{\pi}\left[0 - \frac{1}{n^2}(-1)^n - 0 - \frac{1}{n^2}(-1)^n\right]$$

$$= 0$$

$$\boxed{\therefore\ a_n = 0}$$

Note

$\cos(\pm n\pi) = (-1)^n$

$\sin(\mp n\pi) = 0$

$$b_n = \frac{1}{L}\int_{-L}^{L} f(x)\sin\left(\frac{n\pi x}{L}\right)dx$$

$$\Rightarrow \frac{1}{\pi}\int_{-\pi}^{\pi}(\pi - x)\sin(nx)\,dx$$

	u	v
$(+)$	$\pi - x$	$\sin nx$
$(-)$	-1	$-\frac{\cos nx}{n}$
	0	$-\frac{\sin nx}{n^2}$

$$= \frac{1}{\pi}\left[-(\pi - x)\frac{\cos nx}{n} - \frac{\sin nx}{n^2}\right]_{-\pi}^{\pi}$$

$$= \frac{1}{\pi}\left[\left\{-(\pi-\pi)\frac{\cos(n\pi)}{n} - \frac{1}{n^2}\sin(n\pi)\right\}\right.$$

$$\left. - \left\{-(\pi+\pi)\frac{1}{n}\cos(-n\pi) - \frac{1}{n^2}\sin(-n\pi)\right\}\right]$$

$$= \frac{1}{\pi}\left(2\pi\frac{1}{n}(-1)^n + \frac{1}{2}\sin(n\pi)\right)$$

$$= \frac{1}{\cancel{\pi}}\,2\cancel{\pi}\cdot\frac{1}{n}(-1)^n$$

$$= \frac{2}{n}(-1)^n$$

$$\therefore b_n = (-1)^n\frac{2}{n}$$

$$f(x) = \frac{a_0}{2} + \sum_{n=1}^{\infty}\left[a_n \cos\frac{n\pi x}{L} + b_n \sin\frac{n\pi x}{L}\right];$$

$$= \frac{2\pi}{2} + \sum_{n=1}^{\infty}\left(0 + \frac{2}{n}(-1)^n \sin(nx)\right)$$

$$= \pi + 2\sum_{n=1}^{\infty} \frac{(-1)^n}{n} \sin(nx)$$

$$= \pi + 2\left(\frac{(-1)^1}{1}\sin x + \frac{(-1)^2}{2}\sin(2x) + \frac{(-1)^3}{3}\sin 3x + \frac{(-1)^4}{4}\sin 4x\right.$$

$$f(x) = \pi - 2\sin x + \frac{2}{2}\sin 2x - \frac{2}{3}\sin 3x + \frac{2}{4}\sin 4x$$

$$\cdots$$

Find the fourier expansion

$$f(x) = x \qquad -1 \le x \le 5$$

<u>Solution:</u> $2L = 6 \therefore L = 3$

$$a_0 = \frac{1}{L}\int_{-1}^{5} f(x)\,dx = \frac{1}{3}\int_{-1}^{5} x\,dx =$$

$$= \frac{1}{3}\left[\frac{x^2}{2}\right]_{-1}^{5} = \frac{1}{3}\left(\frac{25}{2} - \frac{1}{2}\right)$$

$$= 4$$

$$a_n = \frac{1}{L}\int_{-L}^{L} \underline{f(x)} \cos\frac{n\pi x}{L}\,dx$$

$$= \frac{1}{3}\int_{-1}^{5} x \cos\frac{n\pi x}{3}\,dx$$

$$= \frac{1}{3}\left[x\frac{3}{n\pi}\sin\frac{n\pi x}{3} + \frac{9}{n^2\pi^2}\cos\frac{n\pi x}{3}\right]_{-1}^{5}$$

u	v
$+x$	$\cos\frac{n\pi x}{3}$
-1	$\frac{3}{n\pi}\sin\frac{n\pi x}{3}$
0	$-\frac{3^2}{n^2\pi^2}\cos\frac{n\pi x}{3}$

$$b_n = \frac{1}{L}\int_{-L}^{L} f(x)\sin\left(\frac{n\pi x}{L}\right)dx$$

Find Fourier series of $\quad 0 \le x \le 8$

$$f(x) = \begin{cases} 2-x & ; \quad 0 < x \le 4 \\ x-6 & ; \quad 4 < x \le 8 \end{cases}$$

<u>solution:</u> $2L = 8 \quad \therefore L = 4$

$$a_0 = \frac{1}{L}\int_{-L}^{L} f(x)\,dx = \frac{1}{4}\int_0^8 f(x)\,dx$$

$$= \frac{1}{2}\int_0^4 f(x)\,dx + \frac{1}{4}\int_4^8 \underline{f(x)}\,dx$$

$$= \frac{1}{2}\int_0^4 2-x\,dx + \frac{1}{4}\int_4^8 x-6\,dx$$

$$\boxed{a_0 = 0}$$

$$a_n = \frac{1}{L}\int_{-L}^{L} f(x)\cos\frac{n\pi x}{L}\,dx$$

$$= \frac{1}{4}\int_0^8 f(x)\cos\frac{n\pi x}{4}\,dx$$

$$= \frac{1}{4}\int_0^4 f(x)\cos\frac{n\pi x}{4}\,dx + \int_0^8 f(x)\cos\frac{n\pi x}{4}\,dx$$

$$= \frac{1}{4}\int_0^4 (2-x)\cos\frac{n\pi x}{4}\,dx + \int_0^8 (x-6)\cos\frac{n\pi x}{4}\,dx$$

$$\boxed{a_n = \frac{8}{n^2\pi^2}\left[1-(-1)^n\right]}$$

$$b_n = \frac{1}{L}\int_{-L}^{L} f(x)\,\sin\frac{n\pi x}{4}\,dx$$

$$\boxed{b_n = 0}$$

$$f(x) = \frac{a_0}{2} + \sum_{n=1}^{\infty} a_n \cos\frac{n\pi x}{4} + b_n \sin\frac{n\pi x}{4}$$

$$b_n = \frac{1}{L}\int_{-L}^{L} f(x)\,\sin\left(\frac{n\pi x}{L}\right)\,dx$$